```
import pandas as pd
import numpy as np
from sklearn.model_selection import train_test_split
from sklearn.linear_model import LogisticRegression
from sklearn.metrics import accuracy_score
```

```
# Load Dataset
df = pd.read_csv("/content/diabetes_prediction_dataset.csv")
```

```
# Display Dataset
print(df.head())
print(df.info())
```

```
   gender   age  hypertension  heart_disease smoking_history    bmi  \
0  Female  80.0             0              1           never  25.19
1  Female  54.0             0              0         No Info  27.32
2    Male  28.0             0              0           never  27.32
3  Female  36.0             0              0         current  23.45
4    Male  76.0             1              1         current  20.14

   HbA1c_level  blood_glucose_level  diabetes
0          6.6                  140         0
1          6.6                   80         0
2          5.7                  158         0
3          5.0                  155         0
4          4.8                  155         0
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 100000 entries, 0 to 99999
Data columns (total 9 columns):
 #   Column               Non-Null Count   Dtype
---  ------               --------------   -----
 0   gender               100000 non-null  object
 1   age                  100000 non-null  float64
 2   hypertension         100000 non-null  int64
 3   heart_disease        100000 non-null  int64
 4   smoking_history      100000 non-null  object
 5   bmi                  100000 non-null  float64
 6   HbA1c_level          100000 non-null  float64
 7   blood_glucose_level  100000 non-null  int64
 8   diabetes             100000 non-null  int64
dtypes: float64(3), int64(4), object(2)
memory usage: 6.9+ MB
None
```

```
# Convert Categorical Columns into Numeric
df["gender"] = df["gender"].map({"Female":0, "Male":1, "Other":2})
df["smoking_history"] = df["smoking_history"].astype("category").cat.codes
```

```
# Features and Target
x = df.drop("diabetes", axis=1)
y = df["diabetes"]
```

```
print("Number of Samples:", x.shape[0])
print("Number of Features:", x.shape[1])
```

```
Number of Samples: 100000
Number of Features: 8
```

```
print("Shape of X:", x.shape)
print("Shape of Y:", y.shape)
```

```
Shape of X: (100000, 8)
Shape of Y: (100000,)
```

```
# Train Test Split
x_train, x_test, y_train, y_test = train_test_split(
    x, y,
    test_size=0.2,
    random_state=0,
    stratify=y
)
```

```
print("Shape of X_train:", x_train.shape)
print("Shape of X_test:", x_test.shape)
```

```
Shape of X_train: (80000, 8)
Shape of X_test: (20000, 8)
```

```
# Logistic Regression Model
model = LogisticRegression(max_iter=500)

model.fit(x_train, y_train)
print("Model Completed")
```

```
Model Completed
```

```
# Prediction
y_pred = model.predict(x_test)

results = pd.DataFrame({
    "Actual": y_test.values,
    "Predicted": y_pred
})

print(results.head(10))
```

```
   Actual  Predicted
0       0          0
1       0          0
2       1          1
3       0          0
4       0          0
5       0          0
6       1          0
```

```
7       0          0
8       0          0
9       0          0
```

```
# Accuracy
accuracy = accuracy_score(y_test, y_pred)
print(f"Test Accuracy: {accuracy:.2%}")
```

```
Test Accuracy: 96.07%
```

```
# New Patient Prediction
new_patient = [[
    1,      # gender (Male=1)
    45,     # age
    0,      # hypertension
    0,      # heart_disease
    2,      # smoking_history
    27.5,   # bmi
    140,    # HbA1c_level
    95      # blood_glucose_level
]]

predicted = model.predict(new_patient)

print("Predicted Diabetes:", predicted[0])
```

```
Predicted Diabetes: 1
/usr/local/lib/python3.13/dist-packages/sklearn/utils/validation.py:2739: Use
  warnings.warn(
```